

Tutorial Sheet 1¹

1. Let $y(x) = \begin{cases} e^x - 1, & x \geq 0 \\ 1 - e^{-x}, & x < 0 \end{cases}$. Check that the derivative of y is continuous. Verify that $y(x)$ is a solution of $y' = |y| + 1$ on $(-\infty, \infty)$.

2. Find the general solution for the following equations.

(a) $y' + 3y = \cos 10x$

(b) $y' + 2y = x^2$

(c) $y' + y - \sin^2 x = 0$

(d) $y' + 2y - (1 + x^3) = 0$

3. Find the general solution for the following equations.

(a) $y' - \frac{2x}{1+x^2}y = 0$

(b) $e^{10x^2}y' - xy = 0$

(c) $(1 + \cos^2 x)y' - \sin 2x y = 0$

(d) $y' + e^{2x} \cos 3x y = 0$

4. Find the general solution for the following equations.

(a) $xy' + 2y = 8x^2$

(b) $(x-2)(x-1)y' - (4x-3)y = (x-2)^3$

(c) $x^2y' + 3xy = e^x$

5. Solve the following non-linear differential equations.

(a) $y' = 2y - 10y^2$

(b) $5x^2y' - 3xy + e^xy^6 = 0$

(c) $xy' + 4y = 16x^2y^{1/2}$

6. Solve the following differential equations.

(a) $\frac{dy}{dx} = \frac{x+3y}{x-y}$

(b) $y' = \frac{x^3 + y^3}{xy^2}$

7. Following may not be separable but can be made separable by substitution.

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$$(a) \ y' = \frac{-6x + y - 3}{2x - y - 1}$$

$$(b) \ y' = \frac{-x + 3y - 14}{x + y - 2}.$$

$$(c) \ (3x + 2y + 2)y' - (2x + 3y + 10) = 0$$

$$(d) \ (x + y - 2)y' - (2x - y - 3) = 0$$

8. Show that the initial value problem $y' = \sqrt{y}$, $y(0) = 0$ has more than one solution by finding at least two solutions explicitly.

9. Find all initial conditions such that $(x^2 - x)y' = (2x - 1)y$ has no solution, precisely one solution, and more than one solution.

10. Let $xy' - 2y = -1$.

(a) Find a general solution to the above problem on $\mathbb{R} - \{0\}$.

(b) Show that y is a general solution for the above ODE if and only if

$$y = \begin{cases} \frac{1}{2} + c_1 x^2, & x \geq 0 \\ \frac{1}{2} + c_2 x^2, & x < 0 \end{cases}$$

where c_1, c_2 are arbitrary constants.

(c) Conclude that all solutions of the ODE on $\mathbb{R}$ are solutions of the initial value problem $xy' - 2y = -1$, $y(0) = \frac{1}{2}$

(d) Show that if $x_0 \neq 0$ and y_0 is arbitrary, then the initial value problem $xy' - 2y = -1$, $y(x_0) = y_0$ has infinitely many solutions on $\mathbb{R}$.

Why does this not contradict existence and uniqueness theorem for linear ODEs?

11. Solve the following IVP's

$$(a) \ (1 + 2y)y' = 2x, \ y(0) = -2.$$

$$(b) \ y' = \frac{(1 + 3x^2)}{3y^2 - 6y}, \ y(0) = 1.$$

$$(c) \ y' = 2 \cos 2x / (3 + 2y), \ y(0) = -1.$$

12. In each of following problems determine (without solving the problem) an interval in which the solution of the given initial value problem is certain to exist.

$$(a) \ y' + (\tan x)y = \sin x, \ y(\pi) = 0.$$

$$(b) \ (4 - x^2)y' + 2xy = 3x^2, \ y(1) = -3.$$

13. In each of following problems solve the given initial value problem and determine how the interval in which the solution exists depends on the initial value y_0 .

(a) $y' + y^3 = 0, \quad y(0) = y_0$

(b) $y' = \frac{x^2}{y(1+x^3)}, y(0) = y_0$

14. (a) Verify that both $y_1(x) = 1 - x$ and $y_2(x) = -x^2/4$ are solutions of the initial value problem

$$y' = \frac{-x + (x^2 + 4y)^{1/2}}{2}, \quad y(2) = -1$$

Where are these solutions valid?

- (b) Explain why the existence of two solutions of the given problem does not contradict the uniqueness part of the existence uniqueness theorem for ODE.

Tutorial Sheet 2

1. Determine if the following equations are exact and solve them.

(a) $(3y \cos x + 4xe^x + 2x^2e^x) dx + (3 \sin x + 3) dy = 0.$

(b) $(\frac{1}{x} + 2x) dx + (\frac{1}{y} + 2y) dy = 0.$

(d) $(ye^{xy} \cos 2x - 2e^{xy} \sin 2x + 2x) dx + (xe^{xy} \cos 2x - 3) dy = 0.$

(e) $\frac{x}{(x^2 + y^2)^{3/2}} dx + \frac{y}{(x^2 + y^2)^{3/2}} dy = 0.$

2. Solve the following IVP.

(a) $(4x^3y^2 - 6x^2y - 2x - 3)dx + (2x^4y - 2x^3) dy = 0 \quad y(1) = 3.$

(b) $(y^3 - 1)e^x dx + 3y^2(e^x + 1) dy = 0, \quad y(0) = 0.$

(c) $(9x^2 + y - 1) dx - (4y - x) dy = 0, \quad y(1) = 0.$

3. Find all the functions M such that the following equation is exact.

$$M(x, y) dx + 2xy \sin x \cos y dy = 0$$

4. Find all the functions N such that the equation is exact.

$$(\ln(xy) + 2y \sin x) dx + N(x, y) dy = 0.$$

5. Suppose M and N are continuous and have continuous partial derivatives M_y and N_x that satisfy the exactness condition $M_y = N_x$ on an open rectangle R around (x_0, y_0) . Show that if (x, y) is in R and

$$F(x, y) = \int_{x_0}^x M(s, y_0) ds + \int_{y_0}^y N(x, t) dt.$$

then $F_x = M$ and $F_y = N$. (HINT: Use Leibniz Rule for differentiation under the integral sign)

6. Solve using the previous exercise. $(x^2 + y^2) dx + 2xy dy = 0.$

7. Solve the initial value problem $y' + \frac{2}{x}y = -\frac{2xy}{x^2 + 2x^2 + 1}, \quad y(1) = -2.$

8. Solve the following after finding an integrating factor.

(a) $(27xy^2 + 8y^3) dx + (18x^2y + 12xy^2) dy = 0.$

(b) $-y dx + (x^4 - x) dy = 0.$

- (c) $y \sin y \, dx + x(\sin y - y \cos y) \, dy = 0.$
- (d) $y(1 + 5 \ln |x|) \, dx + 4x \ln |x| \, dy = 0.$
- (e) $(3x^2y^3 - y^2 + y) \, dx + (-xy + 2x) \, dy = 0.$
- (f) $y \, dx + (2x - ye^y) \, dy = 0.$
- (g) $(a \cos(xy) - y \sin(xy)) \, dx + (b \cos(xy) - x \sin(xy)) \, dy = 0.$

9. Let $y' + p(x)y = f(x)$. Show that $\mu = \pm e^{\int p(x) \, dx}$ is an integrating factor. Find the explicit solution using this integrating factor.

10. Show that if $(N_x - M_y)/(xM - yN) = R$, where R depends on the quantity xy only, then the differential equation $M + Ny' = 0$ has an integrating factor of the form $\mu(xy)$. Find a general formula for this integrating factor.

11. Use the previous problem to solve $(3x + \frac{6}{y}) + (\frac{x^2}{y} + 3\frac{y}{x})\frac{dy}{dx} = 0.$

12. Consider the initial value problem $y' = y^{1/3}, y(0) = 0.$

- (a) Is there a solution that passes through the point $(1, 1)$? If so, find it.
- (b) Is there a solution that passes through the point $(2, 1)$? If so, find it.

13. Apply the Picard's iteration method to the following initial value problems and get four iterations:

- (a) $y' = x + y, y(0) = 0$
- (b) $y' = 2y^2, y(0) = 1$
- (c) $y' = 2\sqrt{y}, y(1) = 0$

Tutorial Sheet 3

1. Find the general solution of $y'' - 2y' + 2y = 0$. Solve it with initial conditions

(a) $y(0) = 3, y'(0) = -2$

(b) $y(0) = k_0, y'(0) = k_1$.

2. Compute the Wronskians of the given set of functions.

(a) $\{e^x, e^x \sin x\}$

(b) $\{x^{1/2}, x^{-1/3}\}$

(c) $\{x \ln |x|, x^2 \ln |x|\}$.

3. Find the Wronskian of a given set of solutions of $y'' + 3(x^2 + 1)y' - 2y = 0$, given that $W(\pi) = 0$.

4. Find the Wronskian of a given set of solutions of $(1 - x^2)y'' - 2xy' + a(a + 1)y = 0$, given that $W(0) = 1$.

5. Find the Wronskian of a given set of solutions of $x^2y'' + xy' + (x^2 - \nu^2)y = 0$, given that $W(1) = 1$.

6. Given one solution y_1 , find other solution y_2 s.t. $\{y_1, y_2\}$ is linearly independent set.

(a) $y'' - 6y' + 9y = 0; y_1 = e^{3x}$,

(b) $x^2y'' - xy' + y = 0; y_1 = x$.

(c) $(x^2 - 4)y'' + 4xy' + 2y = 0; y_1 = 1/(x - 2)$.

7. Suppose p_1, p_2, q_1, q_2 are continuous on (a, b) and the equations $y'' + p_1(x)y' + q_1(x)y = 0$ and $y'' + p_2(x)y' + q_2(x)y = 0$ have the same solutions on (a, b) . Show that $p_1 = p_2$ and $q_1 = q_2$ on (a, b) . [Hint. Use Abel's formula.]

8. Find a linear homogeneous ODE for which the given functions form a fundamental set of solutions on some interval.

(a) $e^x \cos 2x, e^x \sin 2x$

(b) x, e^{2x}

(c) $\cos(\ln x), \sin(\ln x)$.

9. Solve the following IVPs.

(a) $y'' + 14y' + 50y = 0, y(0) = 2, y'(0) = -17$.

(b) $6y'' - y' - y = 0, y(0) = 10, y'(0) = 0$.

(c) $4y'' - 4y' - 3y = 0, y(0) = \frac{13}{12}, y'(0) = \frac{23}{24}$

(d) $4y'' - 12y' + 9y = 0$, $y(0) = 3$, $y'(0) = \frac{5}{2}$

10. Find a particular solution of $x^2y'' + xy' - 4y = 2x^4$.

11. (Principle of Superposition) Assume y_1 is a solution of $a(x)y'' + b(x)y' + c(x)y = f_1(x)$ and y_2 is a solution of $a(x)y'' + b(x)y' + c(x)y = f_2(x)$. Show that $y_1 + y_2$ is a solution of $a(x)y'' + b(x)y' + c(x)y = f_1(x) + f_2(x)$.

12. Find the general solution of

(a) $x^2y'' - 3xy' + 3y = x$

(b) $y'' - 3y' + 2y = 1/(1 + e^{-x})$

(c) $x^2y'' + xy' - 4y = -6x - 4$

(d) $x^2y'' - 2xy' + 2y = x^{9/2}$

(e) $y'' - 2y' + y = 14x^{3/2}e^x$

(f) $y'' + 4xy' + (4x^2 + 2)y = 4e^{-x(x+2)}$, given that $y_1 = e^{-x^2}$, $y_2 = xe^{-x^2}$ are solutions of homogeneous part.

Tutorial Sheet 4

1. Solve the following differential equations

(a) $y''' - y = 0$. $(D^3 - 1) = (D - 1)(D^2 + D + 1)$

(b) $y^{(4)} + 64y = 0$. $(D^4 + 64) = (D^2 + 8 + 40)(D^2 + 8 - 40)$

(c) $y^{(5)} + y^{(4)} + y''' + y'' + y' + y = 0$. $(D + 1)(D^4 + D^3 + 1) = (D^2 + 1 - D)(D^2 + 1 + D)$

(d) $y''' - 2y'' + 4y' - 8y = 0$, $y(0) = 0$, $y'(0) = -2$, $y''(0) = 0$ $(D - 2)(D^2 + 4)$

(e) $y''' - 6y'' + 12y' - 8y = 0$, $y(0) = 1$, $y'(0) = -1$, $y''(0) = -4$ $(D - 2)^3$

(f) $y^{(4)} + 2y''' - 2y'' - 8y' - 8y = 0$, $y(0) = 5$, $y'(0) = -2$, $y''(0) = 6$, $y'''(0) = 8$. $(D + 1)(D - 1)(D^2 + D + \frac{1}{2})$

(g) $y^{(4)} + 2y'' + y = 0$. $(D^4 + 2D^2 + 1) = (D^2 + 1)^2$

2. Find the fundamental set of solutions for the following equations.

(a) $(D^2 + 9)^3 D^2 y = 0$.

(b) $D^3(D - 2)^2(D^2 + 4)^2 y = 0$.

(c) $[(D - 1)^4 - 16]y = 0$

3. Find a particular solution using Anihilator method. Write down the Anihilator explicitly.

Do not evaluate the coefficients.

(a) $y''' - 2y'' + y' = t^3 + 2e^t$ $D^3(D - 1)$

(b) $y^{(4)} - y''' - y'' + y' = t^2 + 4 + t \sin t$. $D^2(D^2 + 1)^2$

(c) $y^{(4)} + 4y'' = \sin 2t + te^t + 4$. $(D - 1)^2 D (D^2 + 4)$

(d) $y''' - 2y'' + y' - 2y = -e^x[(9 - 5x + 4x^2) \cos 2x - (6 - 5x - 3x^2) \sin 2x]$ $(D^2 - 2D + 5)^3$

(e) $y^{(4)} - 7y''' + 18y'' - 20y' + 8y = e^{2x}(3 - 8x - 5x^2)$. $(D - 2)^3$

(f) $y^{(4)} + 5y''' + 9y'' + 7y' + 2y = e^{-x}(30 + 24x) - e^{-2x}$. $(D + 1)^2 (D + 2)$

4. Find the general solution using the annihilator method (method of undetermined coefficients).

(a) $y'' - 2y' - 3y = e^x(-8 + 3x)$.

(b) $y'' + y = e^{-x}(2 - 4x + 2x^2) + e^{3x}(8 - 12x - 10x^2)$.

(c) $y'' + 3y' - 2y = e^{-2x}[(4 + 20x) \cos 3x + (26 - 32x) \sin 3x]$.

(d) $y'' + 2y' + y = 8x^2 \cos x - 4x \sin x$.

(e) $y''' - y'' - y' + y = 2e^{-t} + 3$

(f) $y^{(4)} - 4y'' = 3t + \cos t$.

(g) $y''' - y'' - y' + y = e^x(7 + 6x)$.

(h) $4y^{(4)} - 11y''' - 9y'' - 2y = -e^x(1 - 6x)$.

(i) $y''' + 3y'' + 4y' + 12y = 8 \cos 2x - 16 \sin 2x$.

(j) $y^{(4)} + 3y''' + 2y'' - 2y' - 4y = -e^{-x}(\cos x - \sin x)$

5. Let $P_0(x)y'' + P_1(x)y' + P_2(x)y = F(x)$. Let y_1 be a solution to the corresponding homogeneous equation. Then making the substitution uy_1 in the differential gives a second order equation of the form $Q_0(x)u'' + Q_1(x)u' = F$. This is really a first order equation in variable $z = u'$ and can be solved using the variation of parameters method. This is called the method of **reduction of order**. Use the method of reduction of order to solve $(2-x)y''' + (2x-3)y'' - xy' + y = 0$ given that $y_1(x) = e^x$ is a solution.

Tutorial Sheet 5

1. Determine if the following improper integrals exist.

(a) $\int_0^\infty (t^2 + 1)^{-1} dt$,

(b) $\int_1^\infty t^{-2} e^t dt$

2. Find the Laplace transform of following functions.

(a) $\cosh t \sin t$

(b) $\cosh^2 t$

(c) $t \sinh 2t$

(d) $\sin(t + \frac{\pi}{4})$

(e) $f(t) = \begin{cases} e^{-t}, & 0 \leq t < 1 \\ e^{-2t}, & t \geq 1 \end{cases}$

(f) $f(t) = \begin{cases} t, & 0 \leq t < 1 \\ 1, & t \geq 1 \end{cases}$

$\cosh t = \frac{e^t + e^{-t}}{2}$
 $\sinh t = \frac{e^t - e^{-t}}{2}$

3. (a) Prove that if $L(f(t)) = F(s)$, then $L(t^k f(t)) = (-1)^k F^{(k)}(s)$.

[Hint. Assume that we can differentiate the integral $\int_0^\infty e^{-st} f(t) dt$ with respect to s under the integral sign.]

(b) Using $L(1) = 1/s$, show that $L(t^n) = \frac{n!}{s^{n+1}}$, n an integer.

4. Show that if f is piecewise continuous and of exponential order, then $\lim_{s \rightarrow \infty} F(s) = 0$.

5. Show that if f is continuous on $[0, \infty)$ and of exponential order $s_0 > 0$, then

$$L\left(\int_0^t f(\tau) d\tau\right) = \frac{1}{s} L(f), \quad s > s_0.$$

6. Suppose f is piecewise continuous and of exponential order, and $\lim_{t \rightarrow 0^+} f(t)$ exists. Show that

$$L\left(\frac{f(t)}{t}\right) = \int_s^\infty F(r) dr.$$

7. Suppose f is piecewise continuous on $[0, \infty)$.

(a) If the integral $g(t) = \int_0^t e^{-s_0 \tau} f(\tau) d\tau$ satisfies the inequality $|g(t)| \leq M$, $t \geq 0$, then f has a Laplace transform $F(s)$ defined for $s > s_0$.

[Hint. Use integration by parts to show that

$$\int_0^T e^{-st} f(t) dt = e^{-(s-s_0)T} g(T) + (s-s_0) \int_0^T e^{-(s-s_0)t} g(t) dt$$

(b) Show that if $L(f)$ exists for $s = s_0$, then it exists for $s > s_0$.

$\int_0^\infty e^{-st} f(t) dt \rightarrow \int_s^\infty F(r) dr \rightarrow \int_0^\infty e^{-(s-s_0)t} g(t) dt$

$$\int e^{-st} \frac{\sin \omega t}{t} dt$$

8. Find the Laplace transform of the following functions.

(a) $\frac{\sin \omega t}{t}, \omega > 0,$

(b) $\frac{e^{at} - e^{bt}}{t}$

(c) $\frac{\cosh t - 1}{t},$

(d) $\frac{\sinh^2 t}{t}.$

9. Suppose f is continuous on $[0, T]$ and $f(t + T) = f(t)$ for all $t \geq 0$. We say f is periodic with period T .

(a) Show that the Laplace transform $L(f)$ is defined for $s > 0$.

(b) Show that

$$F(s) = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt, \quad s > 0$$

10. Find the Laplace transform of the following periodic functions.

(a) $f(t) = \begin{cases} t, & 0 \leq t < 1 \\ 2 - t, & 1 \leq t < 2 \end{cases}, \quad f(t + 2) = f(t), \quad t \geq 0.$

(b) $f(t) = \begin{cases} 1, & 0 \leq t < 1/2 \\ -1, & 1/2 \leq t < 1 \end{cases}, \quad f(t + 1) = f(t), \quad t \geq 0.$

(c) $f(t) = \begin{cases} \sin t, & 0 \leq t < \pi \\ 0, & \pi \leq t < 2\pi \end{cases}, \quad f(t + 2\pi) = f(t), \quad t \geq 0.$

(d) $f(t) = |\sin t|.$

11. Find the inverse Laplace transform of the following functions.

(a) $\frac{3}{(s - 7)^4},$

(b) $\frac{2s - 4}{s^2 - 4s + 13},$

(c) $\frac{s^2 - 1}{(s^2 + 1)^2},$

(d) $\frac{s^2 - 4s + 3}{(s^2 - 4s + 5)^2},$

(e) $\frac{s^3 + 2s^2 - s - 3}{(s + 1)^4},$

(f) $\frac{3 - (s + 1)(s - 2)}{(s + 1)(s + 2)(s - 2)},$

(g) $\frac{3 + (s - 2)(10 - 2s - s^2)}{(s - 2)(s + 2)(s - 1)(s + 3)},$

$$(s+1)^3 = s^3 + 3s^2 + 3s + 1$$

$$(s+1)^3 = s^3 + 3s^2 + 3s + 1$$

$$\frac{1}{(s+1)^4} = \frac{(s+1)^2}{(s+1)^4}$$

$$= \frac{a}{s+2} + \frac{b}{s-1} + \frac{c}{s+3}$$

$$\frac{n!}{s^n} \rightarrow \frac{t^{n-1}}{s^n}$$

$$\frac{s^2 + 2s - 10}{(s+1)(s-1)(s+3)}$$

~~(h)~~ $\frac{2+3s}{(s^2+1)(s+2)(s+1)},$

~~(i)~~ $\frac{3s+2}{(s^2+4)(s^2+9)},$

(j) $\frac{17s-15}{(s^2-2s+5)(s^2+2s+10)},$

(k) $\frac{2s+1}{(s^2+1)(s-1)(s-3)}.$

~~12.~~ Solve the following IVP's using Laplace transforms.

~~(a)~~ $y'' + 3y' + 2y = e^t, \quad y(0) = 1, \quad y'(0) = -6,$

~~(b)~~ $y'' - 3y' + 2y = 2e^{3t}, \quad y(0) = 1, \quad y'(0) = -1$

~~(c)~~ $y'' + y = \sin 2t, \quad y(0) = 0, \quad y'(0) = 1,$

~~(d)~~ $y'' + 4y = 3 \sin t, \quad y(0) = 1, \quad y'(0) = -1.$

~~(e)~~ $y'' + y = t, \quad y(0) = 0, \quad y'(0) = 2,$

~~(f)~~ $y'' + 2y' + y = 6 \sin t - 4 \cos t, \quad y(0) = -1, \quad y'(0) = 1.$

~~(g)~~ $y'' - 5y' + 6y = 10e^t \cos t, \quad y(0) = 2, \quad y'(0) = 1,$

~~(h)~~ $y'' + 4y' + 5y = e^{-t}(\cos t + 3 \sin t), \quad y(0) = 0, \quad y'(0) = 4.$

$$L(u(t)) = \frac{e^{-as}}{s}$$

$$L(v(t-a)b(t-a)) = (e^{-as}) L(\underline{b(t)})$$

- $$te^t + te^t(t-1)e^{t-1} + e^{\frac{1}{s-1}}e^{-1}$$

$\alpha = \pi$

$$\frac{e^{-\tau S} (1-2S)}{(S+\nu) + 1}$$

- $$\frac{5-25}{(5-25)^2 + 1}$$

$$(b) \quad H(s) = \frac{1}{s} - \frac{2}{s^3} + e^{-2s} \left(\frac{3}{s} - \frac{1}{s^2} \right) + e^{-3s} \left(\frac{4}{s} + \frac{3}{s^2} \right).$$

- $$(a) \quad y'' - y = \begin{cases} e^{2t}, & 0 \leq t < 2 \\ 1, & t \geq 2 \end{cases} \quad y(0) = 3, \quad y'(0) = -1.$$

$\dot{e}^{2t} = 0$

$$e^{2t} (5 \cos t - 2) = 0.$$

$$(e) \quad y'' - 3y' + 2y = \begin{cases} 0, & 0 \leq t < 2 \\ 2t - 4, & t \geq 2 \end{cases} \quad y(0) = 0, \quad y'(0) = 0.$$

$$(g) \quad y'' + 2y' + 2y = \begin{cases} t^2, & 0 \leq t < 1 \\ -t, & 1 \leq t < 2 \\ -1, & t \geq 3\pi \end{cases} \quad y(0) = 2, \quad y'(0) = -1.$$

- $$c + b =$$

$$f(t) = m + 1, \quad m\pi \leq t < (m+1)\pi, \quad m = 0, 1, \dots$$

$$= 0, 1, \dots$$

$$\frac{1}{e^{\pi}} + \frac{1}{e^{2\pi}}$$

$$1 + \cancel{abc} + \cancel{ac} + bS$$

$$S = 1 + \frac{2}{e^{\pi}} + \frac{3}{e^{2\pi}}$$

$$y'' + y = \begin{matrix} 1 & (0\pi) \\ 2 & (\pi/2\pi) \\ 3 & (\pi\pi) \end{matrix}$$

$$\frac{y}{e^n} = \frac{1}{0} + \frac{1}{e^n} + \frac{y}{e^{2n}}$$

A hand-drawn diagram illustrating the mapping of points from the complex plane to another space. On the left, four red dots are arranged vertically along a vertical line, labeled from top to bottom as $z=0$, $z=1$, $z=2$, and $z=3$. To the right of these dots, there is a large red oval containing a smaller red circle. Further to the right, three more red dots are shown, each associated with a handwritten expression: the top dot is labeled $y(\pi) = 2$, the middle dot is labeled $y'(\pi) \neq 0$, and the bottom dot is labeled $(1 - \frac{1}{e^\pi})$.

- (b) $y'' + y = f(t)$ $y(0) = 0$, $y'(0) = 0$;
 $f(t) = (-1)^m$, $m\pi \leq t < (m+1)\pi$, $m = 0, 1, \dots$
 (c) $y'' - y = f(t)$ $y(0) = 0$, $y'(0) = 0$;
 $f(t) = m + 1$, $m\pi \leq t < (m+1)\pi$, $m = 0, 1, \dots$

Hint: You will need the formula for $1 + r + \dots + r^m = \frac{1 - r^{m+1}}{1 - r}$ ($r \neq 1$).

- (d) $y'' + 2y' + 2y = f(t)$ $y(0) = 0$, $y'(0) = 0$;
 $f(t) = (m+1)(\sin t + 2 \cos t)$, $2m\pi \leq t < 2(m+1)\pi$, $m = 0, 1, \dots$

5. Express the following inverse transform as an integral.

(a) $\frac{1}{s^2(s^2 + 4)}$

(b) $\frac{s}{s^2(s^2 + 4)}$

(c) $\frac{s+2-2}{(s+2)(s^2+9)}$

(d) $\frac{1}{(s+1)^2(s^2+4s+5)}$

(e) $\frac{1}{s^2(s-2)^3}$

6. Find the Laplace transform

(a) $\int_0^t \sin a\tau \cos b(t-\tau) d\tau$.

(b) $\int_0^t \sinh a\tau \cosh b(t-\tau) d\tau$.

(c) $e^t \int_0^t \sin \omega\tau \cos \omega(t-\tau) d\tau$.

(d) $e^t \int_0^t e^{2\tau} \sinh(t-\tau) d\tau$.

(e) $\int_0^t (t-\tau)^4 \sin 2\tau d\tau$.

(f) $\int_0^t (t-\tau)^7 e^{-\tau} \sin 2\tau d\tau$.

(g) $\int_0^t (t-\tau)^7 \tau^8 d\tau$

(h) $\int_0^t (t-\tau)^6 \tau^7 d\tau$

(i) $\int_0^t e^{-\tau} \sin(t-\tau) d\tau$

7. Find a formula for the solutions of the IVP.

(a) $y'' + 3y' + y = f(t)$, $y(0) = 0$, $y'(0) = 0$.

(b) $y'' + 4y = f(t)$, $y(0) = 0$, $y'(0) = 0$.

(c) $y'' + 6y' + 9y = f(t)$, $y(0) = 0$, $y'(0) = -2$.

(d) $y'' + \omega^2 y = f(t)$, $y(0) = a$, $y'(0) = b$.

(e) $y'' - 5y' + 6y = f(t)$, $y(0) = 1$, $y'(0) = 3$.

8. Solve the integral equation

(a) $y(t) = t - \int_0^t (t - \tau)y(\tau) d\tau.$

(b) $y(t) = 1 + 2 \int_0^t \cos(t - \tau)y(\tau) d\tau.$

(c) $y(t) = t + \int_0^t y(\tau)e^{-(t-\tau)} d\tau.$

9. Show that $f * g = g * f.$

10. Show that if $p(s) = as^2 + bs + c$ has distinct real zeros r_1 and r_2 then the solution of

$$ay'' + by' + cy = f(t), \quad y(0) = k_0, \quad y'(0) = k_1$$

is

$$y(t) = k_0 \frac{r_2 e^{r_1 t} - r_2 e^{r_2 t}}{r_2 - r_1} + k_1 \frac{e^{r_2 t} - e^{r_1 t}}{r_2 - r_1} + \frac{1}{a(r_2 - r_1)} \int_0^t (e^{r_2 \tau} - e^{r_1 \tau}) f(t - \tau) d\tau$$

11. For the above problem find a formula for the solution if the roots of $p(s)$ are repeated and is given by r , and when the roots are complex $\lambda \pm i\omega$.

$$a(s^2 L(t) - s K_0 - K_1) + b(s L(t) - K_0) + c L(t) = f(t)$$

$$\frac{f(t) + bK_0 + a s K_0 + a K_1}{c + b s + a s^2}$$